

'Reproducible' Wound Sensing and Trajectory Planning for an Autonomous Wound Care Robot

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Abstract

Automated wound care has the potential to improve hygiene, consistency, and efficiency in clinical practice, yet presents considerable technical challenges in perception and safe manipulation. This dissertation proposes an end-to-end, reproducible pipeline for vision-guided peri-wound cleaning using a Kinova Gen3 7-DOF robotic manipulator. The system integrates DeepSkin-based semantic segmentation with parameterized spiral trajectory generation and robust inverse kinematics solved through TRAC-IK, validated within a ROS 2 simulation environment. Experimental evaluation includes both segmentation repeatability and trajectory sensitivity to initial joint seed configuration, identifying a unique collision-free seed that consistently enables safe spiral execution with the end-effector mounted. All software, configuration files, and datasets are made publicly available to support full reproducibility. Results demonstrate that the method enables repeatable, geometry-aware wound coverage, and highlight key challenges such as peri-wound label variability and motion planning dependence on joint initialization, guiding directions for future research in robust, adaptive deployment.

Declaration of originality

I hereby confirm that no portion of the work referred to in the thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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1 Introduction

Chronic and acute wounds place heavy demands on healthcare, requiring frequent cleaning and monitoring [1]. Manual care is labour-intensive and inconsistent, with outcomes tied to clinician skill and available time. This work develops a fully *reproducible* pipeline, releasing source code, data, and configurations to address reproducibility gaps in autonomous wound care research. The pipeline covers wound segmentation, safe trajectory generation, and execution on a 7-DoF manipulator. We employ **DeepSkin** for multi-class segmentation [2], generate peri-wound spiral trajectories, map them to joint space with **TRAC-IK** [3], and validate execution in ROS 2 with the Kinova Gen3 [4]. Beyond functionality, we analyze segmentation repeatability and planner seed sensitivity.

1.1 Aims and Objectives

Aim. Create a reproducible pipeline that converts wound images into safe peri-wound cleaning trajectories on the Kinova Gen3. **Objectives.** Key steps are: (i) segmenting wound, peri-wound, and background with DeepSkin and deriving peri-wound masks; (ii) parameterizing spiral trajectories from image features; (iii) mapping waypoints to joint space with TRAC-IK and executing them as ROS 2 `JointTrajectory`; (iv) testing segmentation repeatability with per-class precision/IoU; (v) evaluating joint seeds to avoid self-collision; and (vi) publishing scripts and figures for reproduction. **Success criteria.** The pipeline must: (i) execute reliably without collision under a documented seed; (ii) show stable class-level segmentation with peri-wound identified as boundary-sensitive; and (iii) provide reproducible end-to-end scripts and results.

1.2 Report Structure

Section 2 reviews relevant research. Section 3 details the pipeline components. Section 4 reports repeatability and seed-sensitivity studies. Section 5 concludes with future directions such as hardware validation and closed-loop feedback.

2 Literature Review

Robotic manipulation in medicine has expanded greatly in the last two decades, from teleoperated surgery to rehabilitation aids [5]. Yet, routine clinical tasks such as wound irrigation remain under-explored. Manual irrigation demands precision and consistency, especially for chronic wounds, but robotic automation could reduce clinician workload, standardize care, and improve treatment outcomes.

This review surveys research relevant to autonomous wound irrigation with the Kinova Gen3 7-DoF manipulator. It covers wound detection and segmentation, 3D reconstruction, spiral trajectory generation, motion planning with TracIKpy, Kinova Gen3 integration with ROS2, simulation environ-

ments, and hydrogen peroxide (H_2O_2) as the irrigant. Each domain is evaluated for adaptation to autonomous, contactless wound care in both simulated and hardware settings.

2.1 Vision-Based Wound Detection and Segmentation

Accurate wound detection and segmentation are critical for autonomous irrigation. Traditional clinical methods such as visual inspection and manual measurement are subjective and time-consuming. Computer vision, particularly convolutional neural networks (CNNs), now enables objective wound assessment from RGB images.

U-Net, originally developed for biomedical segmentation, remains a backbone for wound pipelines [6], performing well even with limited data. Mishra et al. [7] extended this through DeepSkin, integrating MobileNet for efficiency and achieving Dice scores above 0.85. These results highlight the readiness of 2D segmentation for robotic perception.

2.2 Spiral Trajectory Design for Medical Irrigation

Trajectory patterns strongly affect irrigation outcomes. Spiral paths, moving from wound center outward, displace contaminants, ensure uniform coverage, and reduce abrupt motions—well suited for compliant manipulators like the Kinova Gen3.

Schob et al. [8] showed spiral paths in robotic waterjet debridement yielded smoother forces and better cleaning than raster paths. Similar benefits appear in robotic painting [9]. Spirals can be defined parametrically and mapped to joint space via inverse kinematics.

2.3 Robotic Motion Planning with TRAC-IK and ROS2

Wound irrigation requires precise, smooth motion in semi-structured settings. TRAC-IK provides faster, more robust inverse kinematics than MoveIt2, supporting spiral trajectories where Cartesian and joint motions must be coordinated.

2.4 Kinova Gen3 and ros2_kortex Integration

The Kinova Gen3, a 7-DoF manipulator, offers redundancy and dexterity for medical tasks. The `ros2_kortex` driver provides joint and Cartesian control, trajectory execution, and safety interlocks [4], [10]. In simulation, Ignition Gazebo mirrors hardware interfaces through URDF/Xacro and controller configs.

2.5 Simulation Tools: Gazebo Ignition and RViz2

Ignition Gazebo simulates Kinova Gen3 on rigid wound surfaces, enabling trajectory planning and irrigation coverage estimation. RGB-D sensors such as Realsense support wound reconstruction

within the same pipeline. RViz2 aids visualization and debugging. While Ignition lacks fluid physics, approximations with particles or textures are possible, with CFD offering advanced options [11].

2.6 Automated Use of Hydrogen Peroxide for Irrigation

Hydrogen peroxide (H_2O_2) provides bactericidal action, disrupting biofilm unlike saline [12]. Low-concentration (3%) solutions are safe and effective for superficial wounds [13], [14]. Its bubbling effect supports syringe or peristaltic pump delivery [15]. In this project, H_2O_2 will be **simulated with visual markers**, with future extensions using sensor feedback.

2.7 Related Robotic Systems

Several prior systems provide foundational insights for this project, demonstrating the feasibility of robotic wound interaction, perception-guided planning, and autonomous cleaning tasks.

Jafari et al. [16] presented a robotic wound-filling system using a Staubli TX90 arm and stereo cameras. Their pipeline involved depth-based surface reconstruction, spiral and raster path generation, and material deposition using a 3D inkjet head. Their high-precision path control aligns with the spiral planning strategy adopted here.

Schob et al. [8] demonstrated this in their robotic waterjet debridement system using the DLR MIRO robot. They used rasterised paths for cleaning but here we explore spiral trajectories for improved coverage and fluid dynamics.

Baili et al. [17] developed StapBot, a surgical robot equipped with local sensing capabilities such as ultrasonic and thermal imaging. Their closed-loop decision framework for wound closure demonstrates the value of real-time local sensing in robotic wound management, suggesting future extensions for dynamic contamination scoring during irrigation.

2.8 Gaps in Literature and Future Research Directions

Autonomous wound irrigation is less studied than other medical robotics tasks, with few systems uniting perception, planning, and execution. Key gaps include reproducibility, adaptive feedback, and the absence of open-source fluid models. Segmentation also remains challenged by lighting, occlusion, and skin tone variation.

Future work should target outcome-driven sensing, adaptive end-effectors, and learning-based trajectory design. Rich community datasets are essential for benchmarking [7]. Optimizing spiral parameters and integrating suction present further opportunities.

This pipeline addresses several gaps via ROS 2, TRAC-IK, and hydrogen peroxide, though simplified fluid models and lack of closed-loop control remain limitations.

3 Methods

The methodology underpinning this study establishes a reproducible framework for *autonomous wound-cleaning trajectory execution* using the Kinova Gen3 7-DOF manipulator. The system tightly couples computer vision, geometric modelling, inverse kinematics (IK), and robot control within a ROS 2 ecosystem. In brief, an input RGB wound image is segmented to isolate clinically meaningful regions (wound and peri-wound). From the resulting masks, geometric features (centroid and radial extent) are extracted and used to parameterise a spiral trajectory that uniformly covers the peri-wound area. Each spiral waypoint is converted to a Cartesian pose with a fixed end-effector orientation appropriate for cleaning, mapped to joint space via TRAC-IK, assembled into a time-parameterised `JointTrajectory`, and executed on a simulated Kinova Gen3 controller. RViz markers support visual verification of perception outputs, path generation, and execution. The architecture is intentionally modular and layered to facilitate ablation, debugging, and future substitution of components. It comprises four principal stages: (i) **vision-based wound characterisation**, (ii) **trajectory generation via geometric modelling**, (iii) **inverse kinematic mapping to robot-specific joint configurations**, and (iv) **simulated execution via the Kinova control stack**. The following subsections elaborate each stage in sufficient detail to enable independent reproduction, with explicit parameter settings and implementation notes drawn from the project code base.

3.1 Vision-Based Wound Characterisation and Feature Extraction

3.1.1 Semantic Segmentation of Clinical Wound Images

The identification of wound regions from images is a prerequisite for defining clinically relevant robotic motion. In this study, wound segmentation was achieved using the **DeepSkin** package [2], which employs deep learning-based models to generate multi-class segmentation masks. The masks classify pixels into wound, peri-wound, and background regions, thereby providing structured input for subsequent geometric analysis. The segmentation process operates on RGB wound images that are typically acquired in unconstrained clinical conditions, often with variable lighting and non-standard orientations. To ensure reliability, DeepSkin includes probabilistic thresholding (parameterised by a tolerance value, set at 0.95 in this work) to produce high-confidence wound boundaries. The resulting wound mask provides a binary representation of the lesion, which is further refined using morphological operations such as hole filling and peri-lesion expansion `get_perilesion_mask`. These operations are critical to mitigating artefacts introduced by irregular wound edges or noisy image acquisition. The segmentation output is a three-channel mask where each channel corresponds to a specific region: wound (red), peri-wound (blue), and skin/background (green). This colour coding is consistently used throughout the work for visualisation and verification purposes, as illustrated in Figure 1. In reality, the multi-class masks created on the image are actually just

black images with values of saturation changing from 0 to 2 differentiating the three classes .Example outputs of the segmentation process are shown in Figure 2, demonstrating the original image alongside the wound and peri-wound overlays.

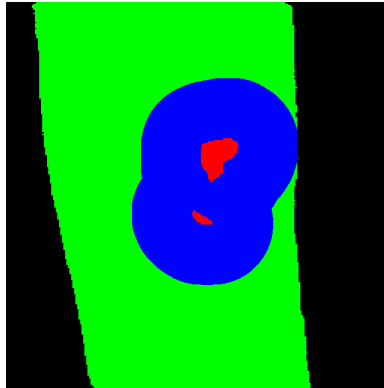


Fig. 1. Segmentation colour coding used throughout this work: wound region shown in red, peri-wound in blue, and skin/background in green.

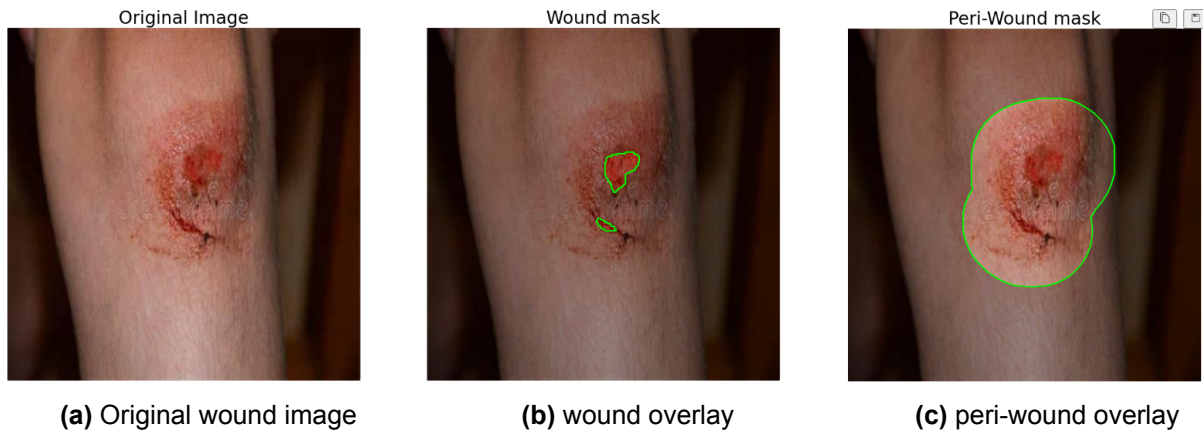


Fig. 2. Comparison of wound image processing steps: (a) original clinical image, (b) image with wound segmentation overlaid, and (c) image with peri-wound region segmentation overlaid.

3.1.2 Geometric Feature Extraction from Segmentation Masks

Once the segmentation masks are obtained, geometric moments are computed to determine the wound centroid. This centroid represents the geometric centre of the wound and serves as the origin for trajectory generation. The peri-wound contour is then analysed to identify the maximum radial distance from the centroid, which is interpreted as the wound radius in pixel space. A calibration factor of 0.1 mm per pixel was applied to convert image-derived distances into metric units. This conversion ensures that the trajectory generation stage operates in a metric scale compatible with the robot's kinematic model. The combination of wound centroid and radial extent provides a clinically relevant geometric abstraction of the wound that can be used for trajectory modelling.

3.2 Mathematical Modelling of Spiral Trajectories

The wound geometry is transformed into a systematic motion pattern by defining a logarithmic spiral centred on the wound centroid. Spiral trajectories are particularly suited to wound-cleaning tasks because they provide uniform coverage while avoiding abrupt tool motion. The spiral was parameterised as:

$$r(\theta) = R \cdot \frac{\theta}{\theta_{\max}}, \quad \theta \in [0, 4\pi], \quad (1)$$

where R denotes the maximum peri-wound radius and $\theta_{\max}$ the angular extent of the spiral.

The spiral was discretised into 50 waypoints, which strike a balance between trajectory smoothness and computational efficiency. Each waypoint was defined in the world coordinate frame of the robot and assigned a fixed vertical offset corresponding to the table height plus 0.0983 m. This offset ensured clearance between the cleaning tool and wound surface, thereby reducing the risk of unintended collision. Visualisation tools in RViz were employed to verify the generated spiral and assess its adherence to the wound geometry.

3.3 Inverse Kinematic Mapping and TRAC-IK Integration

3.3.1 Principles of Inverse Kinematics in Redundant Manipulators

Inverse kinematics (IK) refers to the process of determining the joint configurations of a robot required to achieve a desired Cartesian pose of the end effector. For the Kinova Gen3, a 7-DOF redundant manipulator, IK is inherently non-unique: multiple joint configurations may correspond to the same end-effector pose. Moreover, constraints such as joint limits and singularities further complicate the problem. Traditional analytic solvers are limited in their applicability to redundant robots due to the complexity of symbolic derivations. As such, numerical solvers have become widely adopted for practical implementations. Numerical methods, however, vary in terms of convergence speed, robustness, and computational overhead. To address these challenges, this work employed the **TRAC-IK solver**, a well-established open-source library that extends the capabilities of the classical KDL (Kinematics and Dynamics Library) solver by incorporating randomised seeding and more robust optimisation strategies [3].

3.3.2 Integration of TRAC-IK with Robot URDF Models

TRAC-IK requires detailed knowledge of the robot's kinematic structure to compute inverse kinematics. This information is provided by the Unified Robot Description Format (URDF), an XML-based representation of the robot's joints, links, and physical properties. For the Kinova Gen3 manipulator, the URDF specifies the seven revolute joints, their kinematic limits, the tool frame, and the base frame. During initialisation, TRAC-IK parses the URDF to construct a kinematic chain representing the robot. This chain encodes both the geometric parameters (e.g., Denavit–Hartenberg-

like transformations) and joint constraints (e.g., minimum and maximum joint angles). By leveraging this representation, TRAC-IK is able to search for feasible joint configurations that achieve the specified Cartesian pose. The integration workflow thus begins by loading the Kinova Gen3 URDF into TRAC-IK at runtime. A reference frame hierarchy is established, with `base_link` of the manipulator defined as the root and `tool_frame` as the end effector. Once initialised, TRAC-IK can be queried with Cartesian poses—defined by position vectors and roll–pitch–yaw orientations—and will return joint configurations consistent with the URDF constraints.

3.3.3 Operational Use of TRAC-IK for Robotic Control

In practice, each waypoint of the spiral trajectory was translated into a Cartesian pose with fixed orientation $\text{roll} = \pi$, $\text{pitch} = 0$, $\text{yaw} = 0$, corresponding to a downward tool alignment. These Cartesian targets were passed to TRAC-IK, which computed the corresponding joint angles. To enhance convergence reliability, a persisted seed strategy was employed: the first successful IK solution obtained was stored and reused as the initial seed for subsequent calls. This reduced discontinuities between successive waypoints and improved numerical stability. For waypoints that proved unreachable under direct computation, a nearest-reachable interpolation method was implemented, whereby the target was gradually shifted towards the previous valid point until a feasible configuration was found. The resulting joint configurations were assembled into a ROS 2 trajectory message, with each point spaced by 0.3 seconds. Before execution, the solver-generated joint ordering was remapped to align with the Kinova controller’s expected configuration. This ensured consistency between the planning and execution layers of the system.

3.4 Robotic Simulation Environment and Control Framework

Robotic execution was validated within a simulated environment representing the Kinova Gen3 manipulator [4]. The manipulator was operated under the joint trajectory controller, a standard ROS 2 control interface that accepts trajectory messages and interpolates them into motor commands. The simulated setup provided a risk-free platform for assessing the feasibility of trajectory execution, while ensuring compatibility with hardware deployment in later stages of research. Trajectories were transmitted to the controller through dedicated ROS 2 topics, and visualisation tools were used to monitor execution. Markers representing wound segmentation outputs, spiral waypoints, and executed joint trajectories were displayed in RViz, enabling real-time qualitative assessment of trajectory smoothness and adherence to clinical constraints. This environment facilitated iterative development and tuning of system parameters.

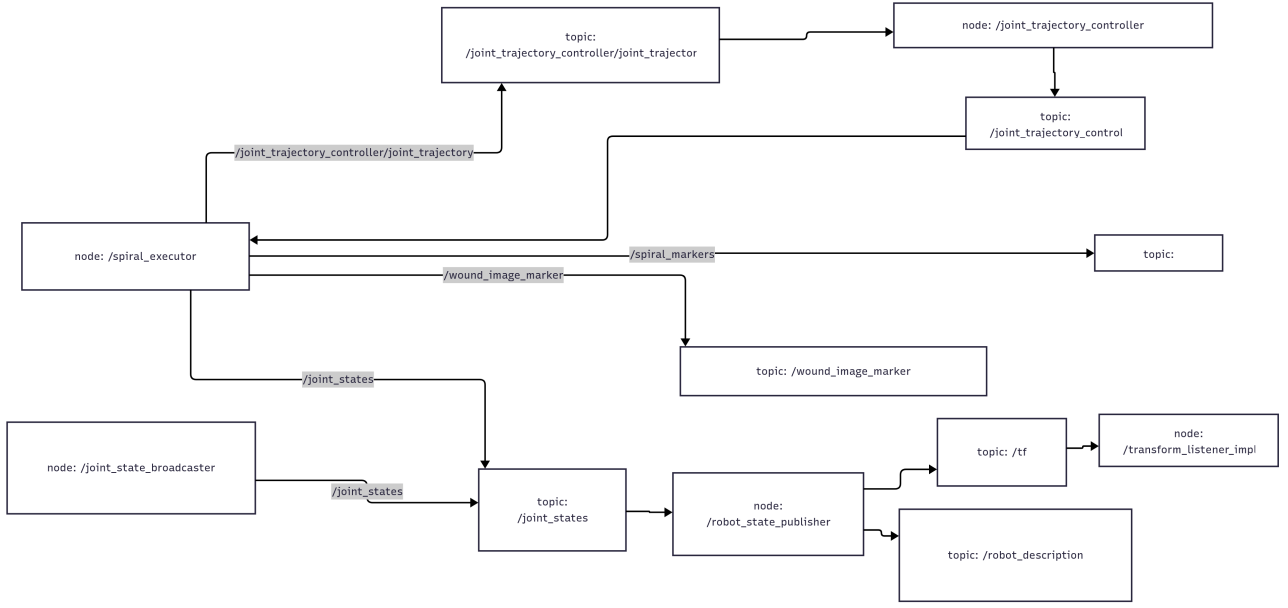


Fig. 3. Graph illustrating the ROS 2 pipeline of the proposed framework

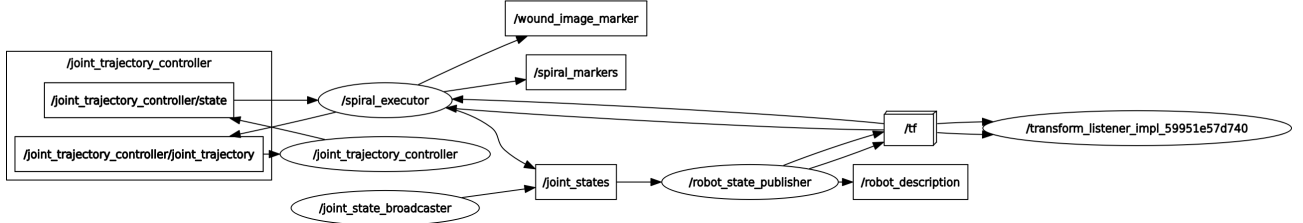


Fig. 4. RQT Graph illustrating the ROS 2 pipeline of the proposed framework

3.5 Methodological Rationale and Design Considerations

The methodological choices were motivated by both engineering feasibility and clinical applicability. By employing DeepSkin for wound segmentation, the system leveraged state-of-the-art computer vision models designed specifically for clinical wounds. The choice of spiral trajectories was informed by the need to ensure uniform coverage while minimising redundant tool paths. The adoption of TRAC-IK reflected a balance between computational tractability and robustness, particularly in the context of redundant manipulators with complex kinematic chains. Moreover, the methodology deliberately emphasised **offline trajectory execution** rather than real-time adaptation. This decision was made to prioritise reproducibility, reduce system complexity, and accommodate the limitations of simulated wound models. By focusing on rigid-surface validation, the system established a robust baseline that can later be extended with tactile sensing and deformable modelling.

3.6 Data Collection and Segmentation Repeatability Protocol

We evaluated run-to-run stability on three smartphone photos (`wound1`–`wound3`). For each image, DeepSkin was run three times at base tolerance 0.95; runs 2–3 used small, deterministic perturbations (tolerance jitter ± 0.02 and optional low-magnitude photometric jitter: brightness/contrast, white

balance, Gaussian noise) with a fixed pseudo-random seed. DeepSkin’s three channels (*wound*, *body/skin*, *background*) were decoded to a label map with priority $\text{wound} > \text{body} > \text{background}$ and IDs $\{0 : \text{background}, 1 : \text{wound}, 2 : \text{body}\}$. A peri-wound mask was derived by dilating the wound (5 px radius), intersecting with body/skin, and excluding the wound, yielding a four-class representation (background, body, wound, peri-wound). All artefacts were saved (raw 3-channel outputs, 3-class labels, peri-wound masks, 4-class labels) using run-indexed names $\langle \text{stem} \rangle_1$, $\langle \text{stem} \rangle_2$, $\langle \text{stem} \rangle_3$. The first run served as reference; runs 2–3 were compared per class using *agreement precision* and *Intersection-over-Union (IoU)*. For each image and class, we report the mean and standard deviation across runs 2–3 to quantify repeatability.

3.7 Summary of Methodological Contributions

Table 1. Summary of system components and their roles within the proposed methodology.

Module	Function	Tool/Library Used
Vision	Wound segmentation and feature extraction	DeepSkin package
Planning	Generation of spiral trajectories covering wound regions	Custom Python (NumPy) implementation
Inverse Kinematics	Mapping of Cartesian waypoints into joint configurations	TRAC-IK solver using URDF model
Control	Execution of trajectories and visualisation	ros2_kortex driver, RViz visualisation

In summary, this methodology integrates vision-based wound segmentation, spiral trajectory generation informed by geometric analysis, and robust inverse kinematic mapping through TRAC-IK. By utilising the robot’s URDF for solver integration, the system ensured kinematic feasibility and compatibility with execution controllers. Validation in simulation demonstrated the viability of the approach, establishing a foundation for future work on hardware validation and adaptive wound-cleaning strategies. This comprehensive pipeline highlights how medical image analysis can be translated into clinically relevant robotic motion. By bridging perception, planning, and control, the methodology represents an important step towards semi-autonomous medical robotics that is accessible to both engineering and clinical communities.

4 Results and Discussion

We evaluate two aspects of the system: (i) the **four-class segmentation repeatability** of DeepSkin (background, body, wound, peri-wound) across repeated runs of the same smartphone images; and (ii) the **trajectory seed sensitivity** of the motion planner (*Q-table seed*). Unless stated otherwise, run #1 serves as the reference; runs #2–#3 are compared against it. We report per-class *precision* and *Intersection over Union (IoU)* as defined in Sec. 3.1.

4.1 Four-Class Segmentation Repeatability

We processed three smartphone wound images. For each image, DeepSkin was executed three times at a base tolerance of 0.95 with minor, controlled perturbations between runs (cf. Methods). We computed per-class IoU and precision for background, body, wound, and peri-wound by treating run #1 as reference. Metrics are summarised visually in Figs. 5 and 6.

4.1.1 Per-Class IoU (Agreement in Extent)

IoU results show **background** and **body** to be the most stable classes across repeated runs, consistent with their broad, low-curvature regions. **Wound** IoU is high but exhibits slightly more dispersion, reflecting sensitivity to lesion-edge placement. The **peri-wound** class shows the largest variability, as it is defined as a thin ring adjacent to the wound; small shifts in wound extent reassign pixels among wound, peri-wound, and body, amplifying disagreement near boundaries (Fig. 5).

4.1.2 Per-Class Precision (Positive Predictive Consistency)

Per-class precision mirrors the IoU trends (Fig. 6). Background/body predictions remain consistently precise across attempts. Wound precision is strong with modest run-to-run spread. Peri-wound precision is the most sensitive to perturbations, confirming that boundary-localised labels are the limiting factor for repeatability.

4.1.3 Implications for the Pipeline

These findings indicate that, for downstream planning, masks derived from background/body/wound are *repeatable* under minor input and parameter variations. Any end-effector trajectories depending on *peri-wound* specifically (e.g., boundary-following or ring-coverage patterns) are more susceptible to small mask changes; stabilisation strategies such as boundary smoothing, morphological consensus across runs, or seed-locked inference can mitigate this.

4.2 Trajectory Planning Seed Sensitivity (Initial Joint Configuration q)

We investigated how the initial joint configuration $q \in \mathbb{R}^7$ (seed) affects the feasibility and stability of the spiral cleaning trajectory when the end-effector (EE) is mounted and a safety standoff from the table is enforced. Each candidate seed was tested with identical waypoints and controller gains; collision checking was enabled in simulation. The objective was to identify a single seed that *consistently* avoids self-collision, respects the EE offset, and executes the full path without posture-induced failure.

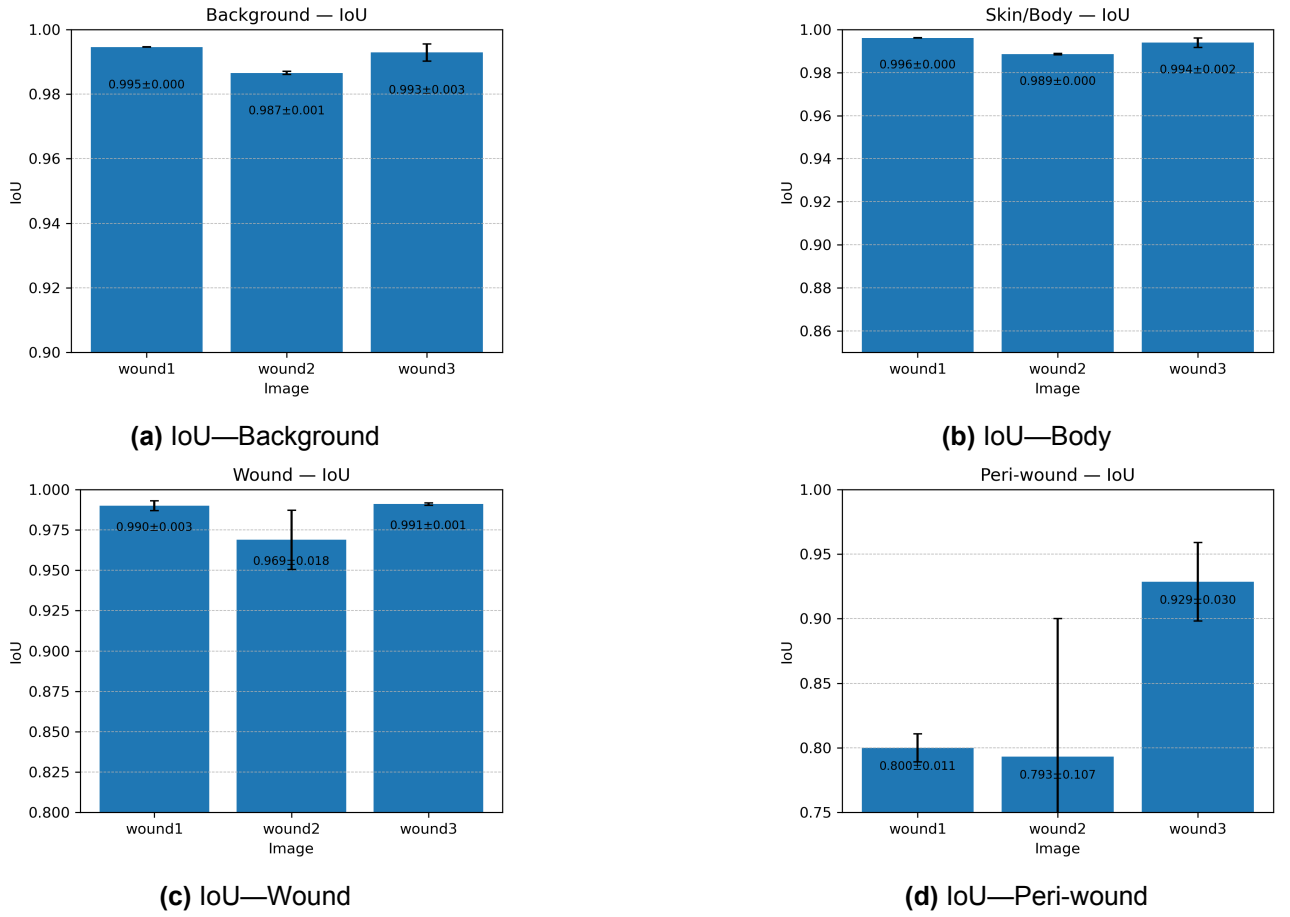


Fig. 5. Per-class IoU across repeated runs (runs #2–#3 vs. run #1).

4.2.1 Candidates and Observed Outcomes

Table 2 summarizes the four evaluated joint seeds (q) along with their qualitative outcomes under varying EE (End Effector) offsets. The evaluated seed joint configurations (in radians) were as fol-

Table 2. Observed Outcomes for Initial Joint Seeds Under EE Offset.

Seed (ID)	Outcome summary
S1	Self-collision when lifting; posture collapses on approach.
S2	Only works with double EE clearance in z ; unreliable at nominal clearance.
S3	Nearly works, but occasionally curls into itself mid-path.
S4 (selected)	Robust: clears table at nominal standoff, no collisions, repeatable spiral completion.

lows:

- **S1:** $[-2.2301, -0.0308, -0.8647, -2.0062, 0.0261, -1.1156, 0.0356]$
- **S2:** $[1.4875, 0.1764, 1.4529, -1.5773, -2.9664, 1.5851, 2.9411]$
- **S3:** $[3.1210, 0.0274, -3.1241, 1.4695, -3.1408, -1.6998, -0.0024]$
- **S4:** $[-1.7837, -0.4392, -1.9867, -1.4959, 0.4121, -1.8122, -0.5628]$

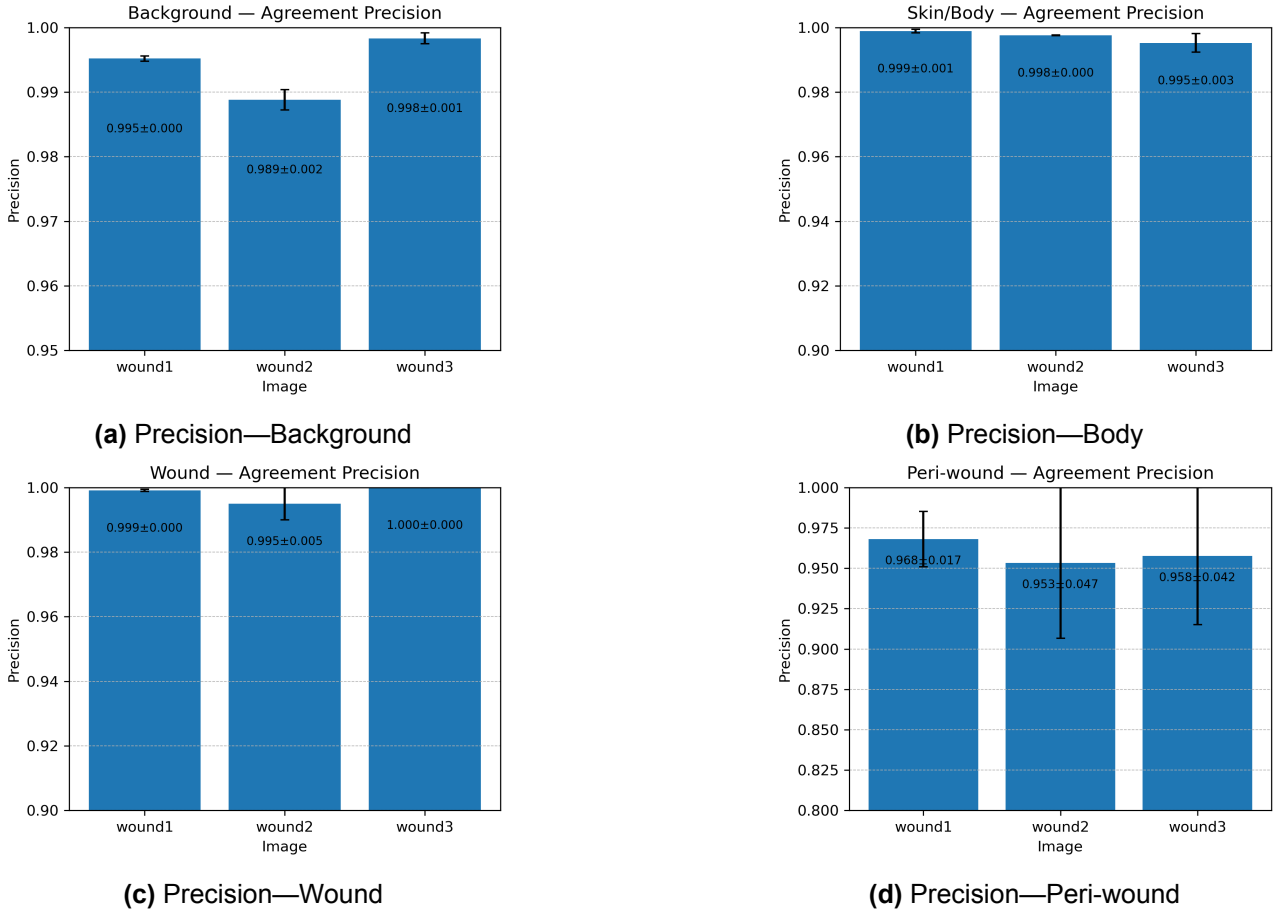


Fig. 6. Per-class precision across repeated runs (runs #2–#3 vs. run #1).

A failed execution under seed S3, where the manipulator folds into itself, is visualized in Appendix C (Fig. 7). For the successful case (seed S4), screenshots from both an orthogonal top view and a side perspective are included (Appendix C.2; Figs. 8a, 8b) to illustrate the spiral trajectory execution without collisions.

4.2.2 Failure Analysis (why S1–S3 underperform)

S1 fails once the EE offset is applied: the initial posture drives an approach that constrains elbow–wrist clearance, so lifting to meet the standoff reorients links into self-collision. **S2** is feasible only with an *over-conservative* standoff (about $2\times$ the EE z), indicating limited clearance margin; with nominal clearance it becomes sensitive to small planning variations. **S3** shows configuration-dependent “curling,” consistent with transiently poor manipulability around the path (elbow/shoulder near a kinematic limit) leading to large null-space motions and link self-approach.

4.2.3 Selected seed and Rationale

S4 is adopted as the *deployment seed* because it: (i) maintains link–link and link–EE clearance under the required standoff; (ii) avoids self-collision across the entire spiral; (iii) executes repeatedly with the tool attached. This posture provides a favorable starting orientation for shoulder–el-

bow–wrist, reducing null-space drift and keeping the forearm away from the upper arm during the surface-following motion.

4.2.4 Summary

Among four plausible seeds, only **S4** satisfies all constraints with the end-effector mounted and nominal table clearance. Earlier seeds either required impractically large standoff or exhibited posture-induced failure (self-collision/curling). We therefore *lock* S4 as the initial configuration for all subsequent trials and hardware-in-the-loop executions.

4.3 Summary of Results

Across repeated runs on the same smartphone images, DeepSkin’s **four-class** segmentation exhibited strong repeatability for *background* and *body*, robust but slightly more variable performance for *wound*, and the greatest variability for *peri-wound*—as expected for a thin, boundary-adjacent label (Figs. 5 and 6). This indicates that downstream behaviours leveraging background/body/wound masks are stable under minor input/parameter perturbations, whereas peri-wound-dependent behaviours (e.g., ring-coverage or boundary-following) are more sensitive and benefit from boundary smoothing, run-consensus, or seed-locked inference for added stability. In motion planning, we evaluated four initial joint seeds (q) with the end-effector attached and a nominal standoff enforced (Table 2). **S1** failed under EE offset due to self-collision during approach; **S2** was feasible only with an over-conservative standoff ($\approx 2 \times \text{EE } z$); **S3** “curled” into itself intermittently, consistent with poor local manipulability; and **S4** satisfied all constraints—no self-collision, nominal clearance, and repeatable execution—and was therefore adopted as the deployment seed. **Implications.**

- *Perception*: Use background/body/wound directly for planning; if peri-wound is required, stabilise boundaries (e.g., morphological smoothing or multi-run consensus) to mitigate ring-label volatility.
- *Planning*: Lock **S4** as the default seed for reproducibility. Where compute allows, sample a small seed set and select the minimum-jerk, collision-free solution to further reduce timing/trajectory variance.
- *System-level*: The combined results support reliable static-path execution; remaining sensitivity is concentrated at wound margins and can be addressed by boundary-aware post-processing or, in future work, closed-loop force/pose feedback.

5 Conclusions and Future Work

5.1 Conclusions

This dissertation presented a reproducible pipeline that converts smartphone wound images into peri-wound cleaning trajectories executed on a simulated Kinova Gen3 platform. The method integrates (i) DeepSkin-based multi-class segmentation with a deterministic peri-wound derivation, (ii) geometric spiral parameterisation from image features (centroid and radial extent), (iii) inverse kinematics via TRAC-IK with seed persistence, and (iv) ROS 2 control and visualisation. Two reliability studies were conducted: segmentation repeatability across repeated runs of identical images and trajectory sensitivity to the initial joint configuration.

Key findings.

- *Segmentation repeatability.* Background and body classes were highly stable across runs; wound remained robust with modest variability; peri-wound exhibited the largest variance, consistent with its boundary-ring construction. Per-class precision and IoU figures corroborate these trends and localise residual instability to wound margins.
- *Trajectory seed sensitivity.* Among four candidate 7-DoF seeds, only one (**S4**) satisfied nominal standoff with the end-effector attached while avoiding self-collision and “curling” throughout the spiral, and was therefore adopted as the deployment seed.
- *System implication.* The pipeline supports reliable static-surface execution when planning derives primarily from background/body/wound masks; behaviours that explicitly depend on peri-wound boundaries benefit from additional stabilisation.

Overall, the results demonstrate that a lightweight perception–planning–control stack can deliver repeatable, geometry-aware trajectories for wound-adjacent coverage using commodity imagery and standard ROS 2 components.

5.2 Future Work

Building on these results, several directions can strengthen robustness, clinical realism, and breadth of validation:

- **Perception robustness.** Increase data diversity and incorporate boundary-aware post-processing (e.g., contour regularisation, multi-run consensus, or uncertainty gating) to stabilise peri-wound labels. Add ground-truth masks to evaluate absolute accuracy (Dice/F1) alongside repeatability.
- **3D awareness.** Introduce depth sensing or photogrammetric reconstruction to project image-space spirals onto non-planar wound surfaces; calibrate pixel–metric scale per scene rather than using a fixed factor.

- **Planning and IK.** Explore task-priority/null-space regulation to reduce elbow–wrist proximity, and compare TRAC-IK with optimisation-based planners under identical tolerances. Automate seed selection (e.g., shortlist candidates, pick minimum-jerk or best-conditioned plan at runtime).
- **Control and safety.** Transition from open-loop to compliant execution (admittance/impedance) with force limits; add online pose correction or visual servoing for drift. Introduce temporal profiles that explicitly bound jerk and tool acceleration.
- **Hardware validation.** Port the pipeline to a physical Gen3 with a sterilizable end-effector, verify standoff and tool alignment, and measure execution metrics (path length, time, peak velocity/jerk) and user-defined safety margins. Conduct pilot evaluations on benchtop phantoms before clinical-grade studies.
- **Clinical integration.** Extend the workflow to include documentation and audit trails (image, mask, trajectory, and execution logs), and assess usability with clinicians for parameter presets and exception handling.

These steps target the two identified bottlenecks—peri-wound boundary stability and seed-dependent execution—toward a deployable, clinically meaningful system that maintains safety while reducing operator burden.

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Appendices

A Project Outline

Title: Autonomous Path Planning on Kinova Gen 3 7-DOF Manipulator for Wound Irrigation Using Computer Vision

Degree: MSc Robotics, University of Manchester

Submitted By: Akshat Taneja

1. Introduction

Robotic wound irrigation offers a compelling alternative to manual methods by enhancing hygiene, consistency, and efficiency in clinical environments. Traditional irrigation techniques, such as syringe flushing or pressurized saline canisters, present several limitations, including inconsistent pressure application and risks of tissue trauma [QxMD Study, Medscape]. This project explores the potential of automating wound irrigation using the Kinova Gen3 7-DOF manipulator combined with computer vision and depth sensing. By enabling autonomous wound detection, segmentation, and targeted irrigation path planning, this system aims to streamline wound care procedures, reduce infection risks, and support clinical staff. The project focuses on surface-level wound cleaning under static conditions, integrating vision-based control through an Intel RealSense RGB-D sensor and evaluating the pipeline within ROS2-based simulation environments.

2. System Architecture and Methodology

The system begins with the acquisition of RGB-D data from a RealSense camera to detect the wound area and extract spatial information. Using deep learning-based methods such as YOLO and FastCNN, along with traditional segmentation as a fallback, wounds are identified and segmented in the image space. To enhance anatomical understanding and segmentation fidelity, auxiliary tools such as SimpleITK [SimpleITK] and reference data from the Visible Human Project [NLM Visible] are incorporated. Wound features—including area, shape, and estimated depth—are derived from point clouds generated from the depth image. These metrics inform the control system on spatial mapping and trajectory generation.

Motion planning is handled via PickNik's MoveIt framework [PickNik], leveraging both joint space and Cartesian space planning strategies. The robot's inverse kinematics, collision avoidance, and workspace constraints are managed through ROS 2 and MoveIt integration. A custom-designed end-effector, inspired by IAAC's airbrush end-effector design [IAAC Airbrush Tool], is planned for deployment as the primary interaction tool with the wound surface, without active fluid delivery. Instead of performing cleaning per se, the focus is on proving the robotic manipulation and trajectory planning pipeline. The system is validated in both simulation using Gazebo and MoveIt and physically using a synthetic arm embedded with wound patterns.

3. Implementation

The project is built on the ROS 2 Humble stack, with core tools including OpenCV for image processing, MoveIt 2 for motion planning, and RViz2 for visualization.

Training of deep learning models is performed using annotated wound datasets on Google Colab, while SimpleITK provides robust preprocessing tools for image filtering and edge enhancement. The Kinova Gen3 simulation is configured via `ros2_kortex`, employing the `joint_trajectory_controller` for accurate motion execution. Coordinate frames and URDF definitions are carefully managed to ensure consistency during motion execution and trajectory planning.

Wound masks are converted into surface-following trajectories, taking into account curvature and spatial orientation. The system executes these paths in both joint and Cartesian spaces, allowing for comparative evaluation. The airbrush-inspired end-effector is integrated into the Kinova model to simulate realistic manipulation conditions, even though active irrigation is not implemented. The goal is to assess the reliability, repeatability, and feasibility of robotic wound interaction in a controlled setting.

4. Results

Quantitative results include precision, recall, and IoU metrics for wound detection and segmentation tasks. Trajectory performance is assessed by comparing simulated and real-world execution, with emphasis on smoothness, wound area coverage, and accuracy of following segmented contours. Qualitative observations from synthetic wound tests are used to assess the physical plausibility and stability of the manipulation pipeline.

5. Discussion

The results demonstrate strong segmentation performance using RGB-D sensing and deep learning, although challenges remain in handling noisy depth data and complex surface curvature. Robotic motion control using joint and Cartesian strategies yielded consistent performance, though stability of path execution and end-effector compliance are areas for improvement. While fluid delivery was not implemented in this version, the work lays a strong foundation for future integration. From a medical robotics perspective, the demonstration of reliable wound-following motion in simulation and on synthetic surfaces offers promising directions for next-generation robotic wound care.

6. Conclusion and Future Work

This dissertation presents a complete pipeline for robotic wound detection and trajectory planning using a Kinova Gen3 manipulator. Simulated and physical experiments validate the system's capability to localize and track wound contours with precision. The project delivers a deployable robot manipulation module (in simulation and/or hardware) as the final result. Future work should focus on integrating real-time feedback control, adding fluid delivery mechanisms, testing on deformable materials, and refining wound segmentation algorithms for improved generalization.

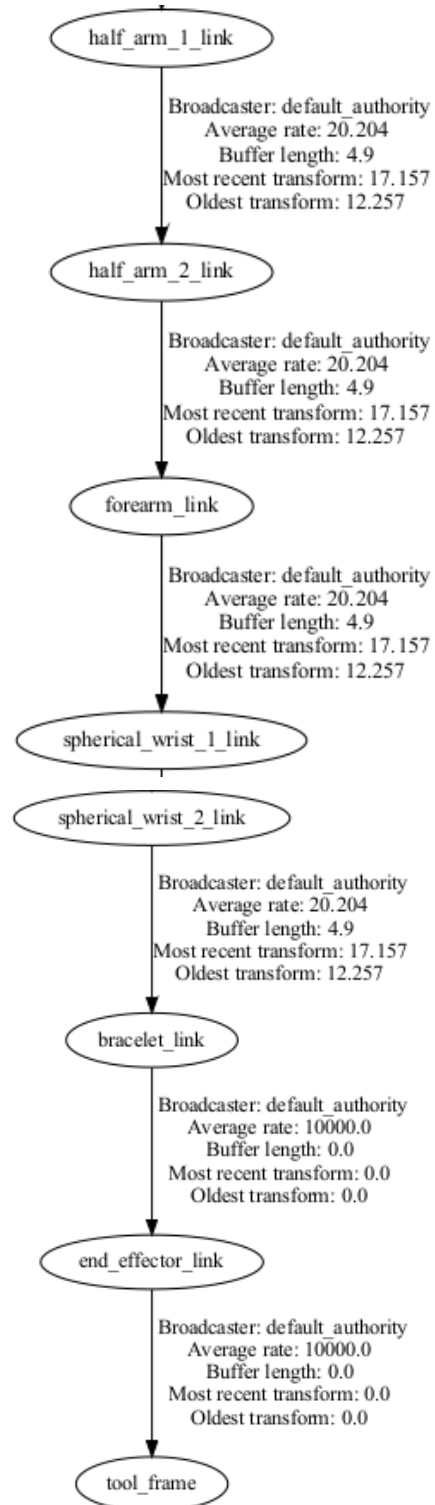
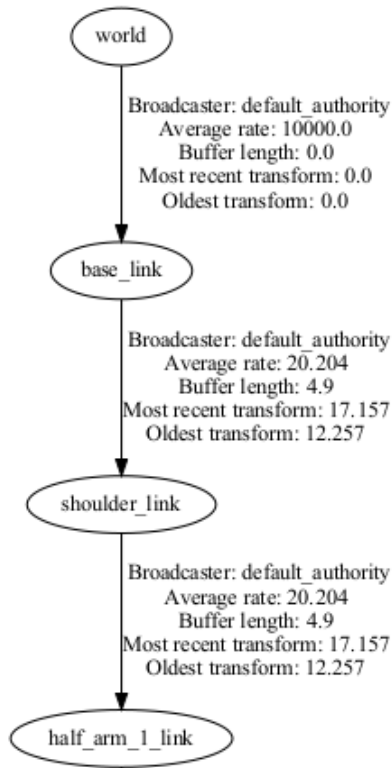
APPENDICES

APPENDIX A

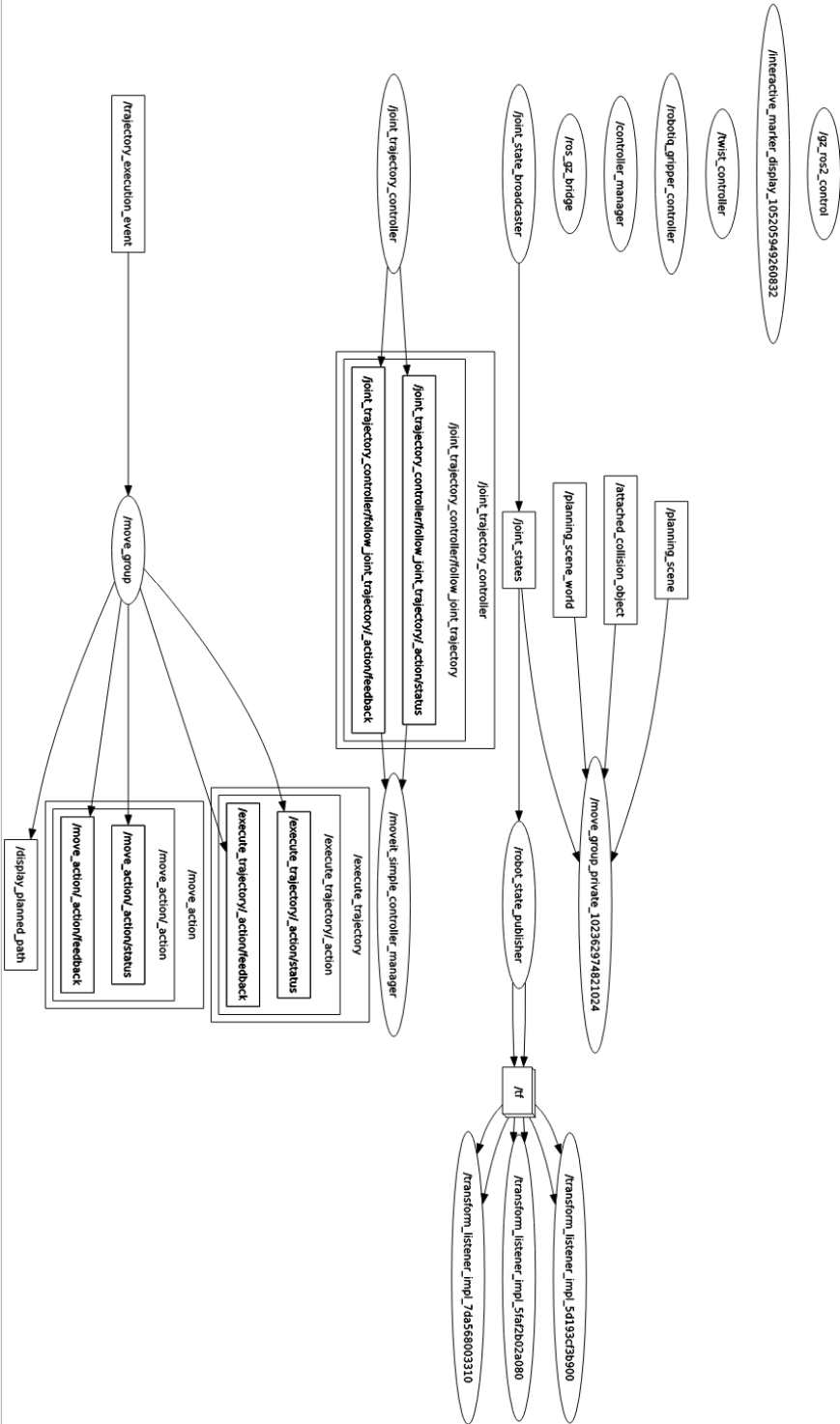
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APPENDIX B (TF GRAPH)



APPENDIX C (RQT GRAPH)



B Risk Assessment

RISK ASSESSMENT

Technical Risks

Risk	Likelihood	Impact	Mitigation Measures
Errors in point cloud reconstruction (e.g., due to occlusions or noise)	Medium	Medium	Use filters, multi-view cameras, and pre-validation with synthetic datasets.
Inaccurate trajectory planning due to poor segmentation	Medium	Medium	Train/test robust segmentation models; validate against reference data.
Collision of manipulator in simulation or during test	Medium	High	Implement collision detection, safety boundaries, and emergency stop logic.
Software bugs or simulation crashes	High	Medium	Use version control, modular development, and automated testing.
Misalignment between point cloud and physical dummy	Medium	Medium	Use registration techniques and validate mapping accuracy manually.
Mechanical failure of robot (e.g., backlash, motor issues)	Low	High	Use rated components; regularly maintain equipment and conduct pre-run checks.
Hardware/software integration issues	Medium	Medium	Modular testing of each component (vision, control, hardware). Reserve time for integration phase.

Practical Risks

Risk	Likelihood	Impact	Mitigation Measures
Limited lab access or equipment failure	Low–Medium	High	Schedule early bookings. Maintain backup plans using virtual or recorded data.

Damage to robotic arm or dummy during testing	Medium	Medium	Use soft end-effectors for testing. Limit joint speeds and torques.
Data loss (e.g., code, logs, 3D reconstructions)	Medium	High	Use Git and cloud backups. Save versions incrementally.
Handling of dummy for positioning/setup	Low	Low	Lightweight models should be used. Always stabilize with clamps or holders.

Health and Safety Risks

Risk	Likelihood	Impact	Mitigation Measures
Minor injuries from manipulator movement	Low	Medium	Operate in supervised, controlled environment. Include e-stop and soft tool tips.
Eye strain or posture issues	High	Low	Take regular breaks. Maintain ergonomic workstation.
Slipping/tripping over cables or testing setup	Low	Medium	Use cable management solutions and secure the workspace.
Electrical issues (shorts, sparks)	Low	High	Inspect all wiring and use protective casings. Never leave powered systems unattended.

Ethical and Legal Risks

Risk	Likelihood	Impact	Mitigation Measures
Misunderstanding that real human tissue is used	Low	High	Clearly state in all documentation and labels that only synthetic, inert models are used.
Inappropriate use or display of anatomical models	Low	Medium	Use commercially available, professional-grade limb analogues. Avoid overly realistic or graphic designs.
Ethical concerns about simulating medical	Low	Medium	Explicitly limit claims to pre-clinical, research-only

robotics without clinical oversight			simulation. Do not propose clinical use or recommendations.
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C Supporting Figures From Seed Sensitivity Study

C.1 Failed Execution Under Seed S3

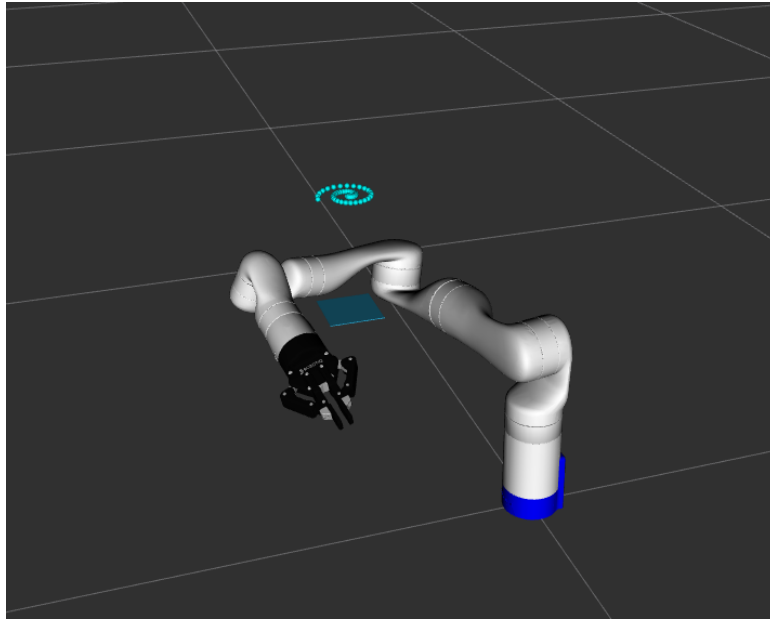
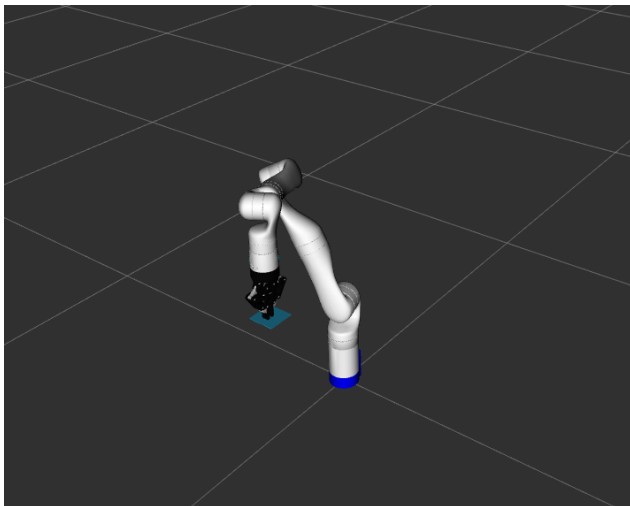
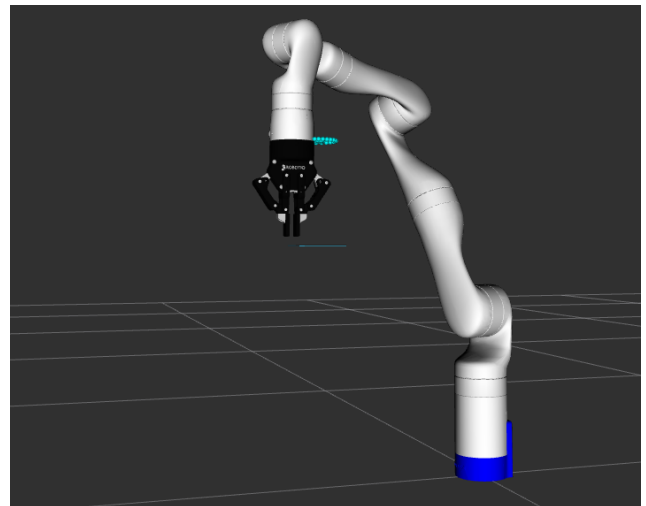


Fig. 7. Failed spiral execution under seed S3, where the manipulator folds into itself

C.2 Successful Execution Under Seed S4



(a) orthogonal view of successful spiral execution under seed S4



(b) side view of successful spiral execution under seed S4

Fig. 8. Successful spiral execution under seed S4, illustrating the end-effector maintaining standoff without collisions

D Other: Code Files And How to Reproduce The Work

The code to reproduce the work in this dissertation is available at: <https://github.com/akshattnj/UoMFinalDissertation>. The repository includes:

- `README.md`: Instructions to set up the ROS 2 workspace
- `scripts/`: Python scripts for segmentation, trajectory generation, IK mapping, and execution
- `images/`: Sample wound images and segmentation masks
- `configs/`: Configuration files for DeepSkin and TRAC-IK
- `results/`: Output logs, precision/IoU calculations, and seed sensitivity data

To reproduce the results:

1. Follow the setup instructions in `README.md` to install ROS 2 Humble and required dependencies.
2. Clone the repository and build the ROS 2 workspace.
3. Run the main pipeline script with a sample image to generate and execute the spiral trajectory.
4. Use the provided analysis scripts to compute segmentation repeatability metrics and visualize seed sensitivity outcomes.

Ethical Considerations

Autonomous robotic systems for wound care, such as the Kinova Gen3 manipulator used in this research, raise several important ethical considerations that must be addressed even at the developmental and preclinical stage. **1. Patient Safety and Harm Prevention**

Ensuring the safety of patients is paramount. Although this work is conducted primarily in simulation, any eventual deployment on humans requires rigorous validation to prevent unintended injury or tissue damage during irrigation and cleaning. The system must include fail-safes, emergency stop mechanisms, and compliance features to minimize risk. **2. Clinical Effectiveness and Reliability**

Ethical obligation mandates that the robotic system performs reliably and consistently under varied clinical conditions. Variability in wound appearance, morphology, and patient anatomy must be managed to avoid suboptimal care or harm caused by automation errors. **3. Data Privacy and Informed Consent**

Use of clinical wound images implies handling sensitive patient data. Strict policies to anonymize, protect, and securely store such data are needed, with patient informed consent for data capture and usage documented according to institutional guidelines. **4. Transparency and Accountability**

Design and deployment of robotic wound care must maintain transparency regarding capabilities and limitations. Accountability mechanisms should define responsibility in case of malfunctions, including clear roles of supervising clinicians and engineers. **5. Social and Accessibility Implications**

Automation aims to reduce clinician workload and improve care consistency; however, ethical deployment includes ensuring equitable access for diverse patient populations without exacerbating healthcare disparities. Special attention must be paid to scoring robustness across different skin tones and wound types. **6. Ethical Use of Simulation and Preclinical Testing**

Use of synthetic models and simulations avoids direct risk to patients and aligns with ethical research standards. Nonetheless, transition to clinical use must follow comprehensive ethical review and regulatory approval processes. This ethical framework guides the responsible development and future clinical translation of autonomous wound irrigation robotic systems, fostering safe, effective, and equitable patient care.